Physics Project - **Research Plan**

<u>Project title:</u> *"Introduction to the Hartree-Fock approximation"*

This project centres around bypassing the inherent difficulty in finding the quantum mechanical behaviours in the many-body problem of multiple interacting quantum particles. Exact solutions can be found from the Schrodinger equation for the simple case of the hydrogen atom with a single bound electron however for atoms with multiple bound and interacting electrons the solution is not so simple, and in this project I will examine the methods for modelling these many-body systems where no exact or explicit solutions.

This work shall have been motivated firstly by a desire to understand and enable the use of methods of finding the quantum behaviours of a broader range of quantum mechanical systems than just those for which the usual equations of quantum mechanics suffices. And secondly this project provides a good starting point for beginning to understand Feynman diagrams and how they can be employed to depict the perturbative interactions, creation, and annihilation of quasi-particles. In this respect it can be thought of secondarily as a starting point to approaching this highly technical diagrammatic representation of quantum processes.

Previously I have undertaken immediately relevant supporting work in a condensed matter physics module and an introduction to quantum mechanics as well as a dedicated quantum mechanics module and an electrodynamics module throughout the three years of my bachelor's degree. What's more I participated in a UROS research project entitled "Modelling quantum mechanics: The ring polymer method" which will also be potentially relevant to this project, however I recognise that the work I undertook in that project dealt with a single non-interacting particle however I shall conduct a short investigation into its reapplication and implementation in the case of the many-body system.

<u>Literature Survey:</u>

Publication Title	Author	Year of Publication
Quantum Mechanics (Volume 1 & 2)	Cohen-Tannoudji, Claude	2005 (2nd Ed.)
Modern Quantum Mechanics	Sakurai, Jun John	2014 (2nd Ed.)
Quantum Mechanics	Rae, Alistair	2016 (6th Ed.)
Molecular Quantum Mechanics	Atkins, Peter William	2011 (5th Ed.)
Schaum's outline of Quantum Mechanics	Peleg, Toav	2010 (2nd Ed.)
Introduction to Quantum Mechanics: with applications to Chemistry	Pauling, Linus	1985
The Feynman lectures on Physics (Volume 2 & 3)	Feynman, Richard Phillips	2011 (New Millen. Ed.)
Energy Levels in Atoms and Molecules	Richard William Graham	1994
A Guide to Feynman Diagrams in the Many-Body Problem	Mattuck, Richard D.	(2nd Ed.)
Quantum Theory of Many-Particle Systems	Fetter, Alexander A. & Walecka, John Dirk	
Quantum Theory of the Electron Liquid	Guiliani, Gabriele F. & Vignale, Giovanni	

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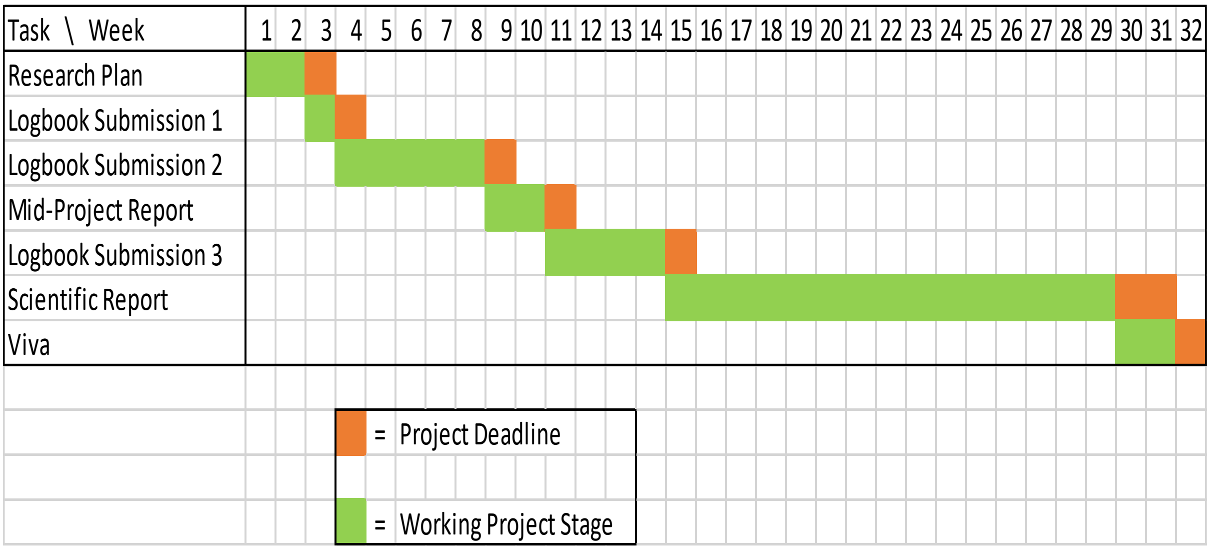
The Oxford: Solid State Basics	Simon, Stephen H.	
Electricity and Magnetism	Purcell, Edward M.	
The Quantum Theory of Atoms Molecules and Photons	Avery	
Molecular Quantum Mechanics	Atkins, Peter & Friedman, Ronald	(5th Ed.)
Quantum Mechanics: non-relativistic theory	Landau, L. D. & Lifshits, E. M.	(3d. Ed.)
Introduction to Quantum Mechanics	Griffiths, David J.	(2nd Ed.)

The equipment and facilities that shall be required as detailed in my purchase requisition shall be quite limited owing to the predominantly theoretical nature of this project, however facilities for scientific computation will at times be required. Where necessary I will implement any necessary computer aided calculations using the computer languages of C++ and Python. A risk assessment form shall also be omitted from this research plan as it is the joint determination of myself and my supervisors that this will not be necessary in this instance or at this stage of the project due owing to the fact that the level of health risk involved in this project does not exceed those of everyday activities and as such falls within the realms of the most basic common sense. It is thus not necessary to explicitly outline any potential health risks with regards to pursuing this research project.

Project Procedure



This Gantt chart details the schedule for the deadlines of this project and the time in weeks that will be spent on each portion. The features of my methodological flowchart shall be grouped according to this schedule as follows:

From the start of the project to the topic core reading and model derivation portion I expect to have completed to the 6th of December, to feature in the discussions of my mid-project report. After this I shall allocate 4 weeks to model refinements and the second occurrence of the simulation/ data analysis section. For the time allocated for the scientific report I allocate the rest of the flowchart; recognising that these flowchart elements will form part of but not the entirety of said report.

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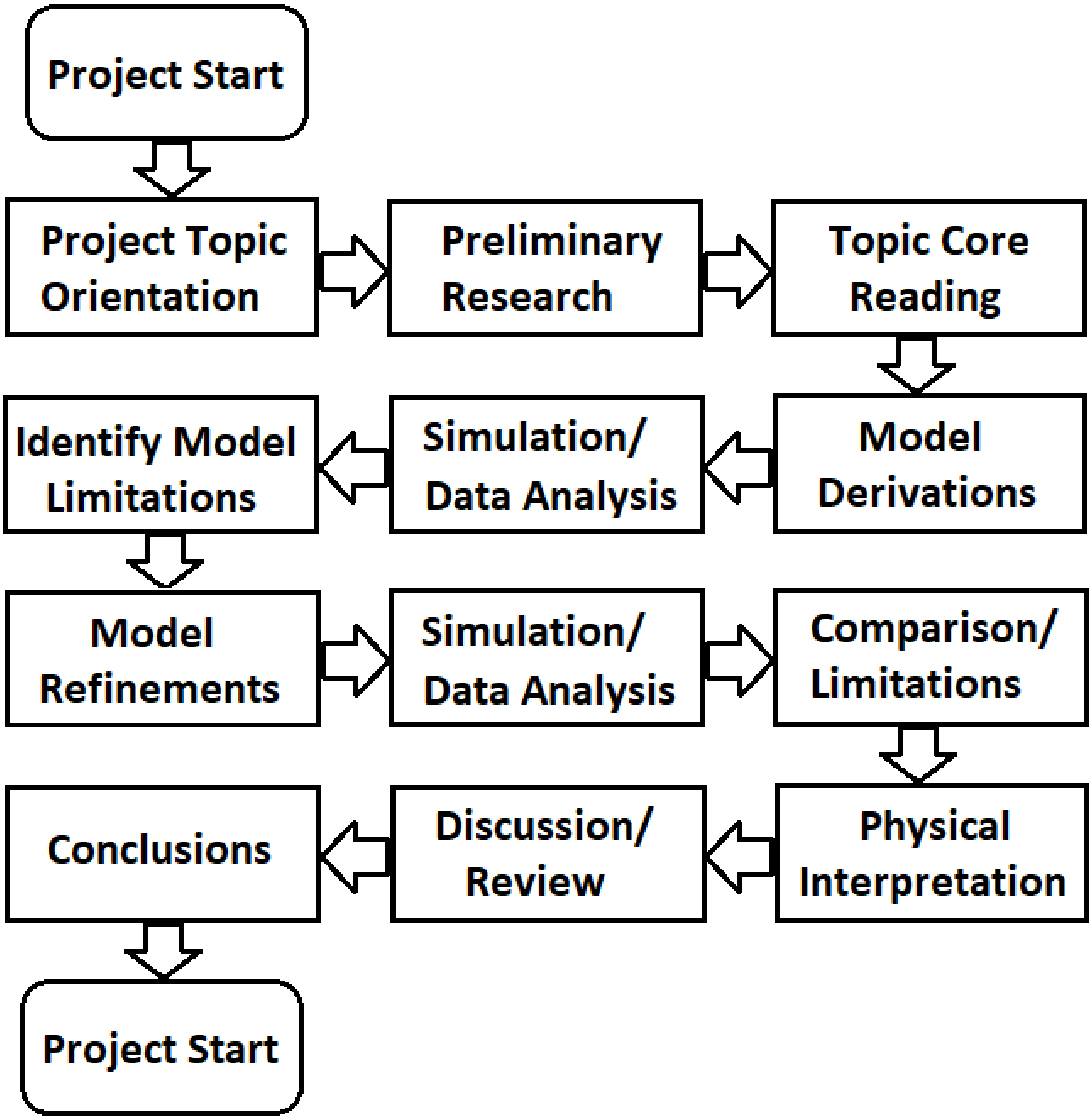
Project Progression Flowchat:



This flowchart outlines the basic activities and set of sequential tasks to be carried out over the course of this project, subject to modification if the circumstances should arise due to better convenience, or to greater understanding of the underlying topic content as it relates to other stages of the research project.

- End of Project plan -

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